Total No. of Questions : 10] [Total No. of Printed Pages : 4

Paper Code : 21317

F-417

B.C.A. (Fourth Semester)

Examination, 2025

(New Course)

Paper -BCA-(N)-402

Introduction to DBMS and SQL

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks. Symbol used have their usual meanings.

1. (a) What is database management system (DBMS)? What are the tasks performed by users in DBMS? What are the advantages and disadvantages of DBMS?

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(b) What is functional dependency? Explain its role in database design. Describe the inference rules for functional dependencies.

2. (a) What is relational model? Explain with example.

(b) Prove that BCNF is stricter than 3NF

3. (a) Who are data administrators? What are the functions of database administrator?

(b) Write short note on SQL. Explain various characteristics of SQL. What are the advantages and disadvantages of SQL.

4. (a) Define normal forms. List the definitions of first, second and third normal forms. Explain BCNF with a suitable example

(b) Discuss the following terms with suitable examples of each.

(i) DDL Command

(ii) DML Command

(iii) DCL

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5. (a) Discuss the architecture of DBMS. What are the type of DBMS architecture?

(b) Describe the classification of database language. What type of language is SQL?

6. (a) What are data models? Briefly explain different types of data models.

(b) Explain the various types of security and integrity threats in a database system. How can these threats impact data reliability?

7. (a) Give an over view of ORACLE architecture and features. How does ORACLE support transaction integrity?

(b) Describe different defense mechanisms used to protect a database system from security threats. Give examples where applicable.

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8. (a) How are constraints and indexes specified in SQl? Discuss the purpose and benefits of using them.

(b) Explain the anomalies that can occur in unnormalized database. How does normalization help eliminate them?

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Paper Code :-21317

F-417

B.C.A. (Fourth Semester)

Examination, 2024

(New Course)

Paper No. : BCA-N-402

Introduction to DBMS and SQL

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Explain the steps of normalization in Brief.

   (b) What is data model? Why is it important. Discuss advantages and disadvantages of data model.

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2. (a) Explain the concept of join Dependency with example in detail.

   (b) Explain Primary key, Foreign key, candidate Key, Super key & Composite key.

3. (a) Explain the different types of defenec mechanism.

   (b) What do you mean by data independence. Explain the difference between logical and physical data independence.

4. (a) Differentiate between DDL and DML in detail.

   (b) What are the anomalies of Third Normal Form (3NF)?

5. (a) What are the anomalies which leads to the bad design of database.

   (b) What is database Security? Why is database security important for an organization.

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6. (a) Discuss the comparison and contrast between a relation in BCNF and 3NF.

(b) Relational Model posses data independence. Explain.

7. (a) What are the features of data in the database? What operations can be performed on a database?

(b) Discuss history of Oracle. What are the various tools provided by the oracle?

8. (a) What is functional dependency? Explain with example.

(b) How do we convert an unnormalized relation to that in 1NF? What are the advantages and disadvantages of Normalization?

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Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21317

F-417

B.C.A. (Fourth Semester) Examination, 2023

(New Course)

Paper No. BCA-N-402

INTRODUCTION TO DBMS AND SQL

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Discuss three level of abstraction or schema architecture of DBMS? 7

   (b) What is the role of database administrator in database management system? Describe weak and strong entity set in database with examples? 7

2. (a) Explain all database languages in detail with examples? 7

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   (b) Explain Data modeling in database. Why would you choose a database system instead of simply storing data in operating system files? 7

3. (a) Explain the Primary Key, Super Key, Foreign Key and Candidate Key with examples? 7

   (b) What is Normalization? How 4NF is differing from PJNF? Explain with examples? 7

4. (a) Consider the following relations : 7

   Hotel {hotelNo, name, address} Room {roomNo, hotelNo, type, price}

   Booking {hotelNo, guestNo, dateFrom, dateTo, roomNo}

   Guest {guestNo, name, address}

   Write the SQL statements for the following:

   (i) List the names and addresses of all guests in Chandigarh, alphabetically ordered by name.

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(ii) List all family rooms with a price below Rs. 400 per night, in ascending order of price.

(iii) How many Hotels are there.

(iv) List of Guest name with Room No in ascending order

(b) Differentiate between Hierarchical model and Network Model? 7

5. (a) How does Boyce-Codd normal form is differ from 3 NF? Why is it considered stronger than 3 NF? 7

(b) What is functional dependency? Discuss its types. Explain the role of functional dependency in the process of normalization. 7

6. (a) What are the arithmetical operators in SQL? Explain with examples? 7

(b) What do you mean by integrity constraints? Explain with examples? 7

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7. (a) Explain Normalization? What are the difference between 4NF and 5NF? 7

(b) What are the different types of anomalies associated with database? Describe with examples? 7

8. Write short notes with suitable examples : 14

(i) RDBMS

(ii) 1NF and 2NF

(iii) Transaction

(iv) Distributed database

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Paper Code : 21317
F–417

B. C. A. (Fourth Semester) EXAMINATION, 2019
(New Course)

Paper No. BCA—N—402

INTRODUCTION TO DBMS AND SQL

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) How many types of data model are for DBMS ? Explain the network model and Hierarchical Model with an example. 7

   (b) Draw system architecture of DBMS. List any *four* significant differences between a life-processing system and a DBMS. 7

2. (a) What is the need of relational model ? Differentiate the relational and non-relational model with example. 7

   (b) Write SQL command to Get Supplier Names for Suppliers who supply all Parts. Using the following tables : 7

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SUPPLIER [S_NO {CHAR(2)primary key}, SPNAME {CHAR (5)}, CITY {CHAR (8)}

PARTS [P_NO {CHAR(2) primary_key}, PNAME {CHARS(5)}, COLOR {CHAR(5)], CITY {CHAR (8)}].

SUP_PARTS [{S_NO CHAR (2), P_NO CHAR (2)} primary_key, QTY NUMBER]

[Where-S_NO:- Supplier number, SNAME:- Supplier Name, P_NO:- Parts Number, PNAME:- Parts Name]

3. (a) What is need of the normalization ? Consider the following universal relation : 7

   $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies :

   $R = \{AB \rightarrow C, A \rightarrow D, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$

   What is the key for R ? Decompose R into 2NF and then 3NF.

   (b) What do you mean by functional dependency ? Describe their applications, what is MVD and join dependency ? 7

4. (a) What is database system ? Explain the elements of database system. 7

   (b) What is the difference between a Database and DBMS ? Explain with example. 7

5. (a) Describe the following : 7

   (i) Data query, manipulation and reporting

   (ii) Concurrent data access for multiple users

   (b) Compare physical and logical models of data. 7

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6. Consider the following relation : 14

Student (Rollno, Name, Subject, Marks)

Write the SQL of the following :

(i) Name of students who have secured the highest marks in the class.

(ii) Name of students who have secured the highest marks in each subject.

(iii) Sort the students in ascending order of their names.

7. (a) What are data models ? Explain advantages and disadvantages of data models. 7

(b) Compare different types of data models. 7

8. (a) What do you mean by integrity constraints ? Explain. 7

(b) Explain Normalisation. Describe 1NF and 2NF and 3NF with example. 7

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